Ritesh Somashekar

AI Engineer | LLM Applications, RAG & Agents

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PROFESSIONAL SUMMARY

- **AI Engineer** with **5+ years** growing from **data engineering** to **machine learning** and **Generative AI**, building **LLM applications, RAG pipelines, and AI agents** in university research and healthcare; published at **ACM WSDM '26** and **IEEE PST 2026**.
- Builds **full-stack AI applications** with **Python, FastAPI, Streamlit, SQL databases, Docker, Kubernetes,** and **CI/CD**, integrating **LLMs, embeddings, and vector databases** into secure, **HIPAA-compliant** systems.
- Proven results: **RAG recall@10 raised from 61% to 88%**, a **3.0% LLM hallucination rate**, **38% faster** support resolution, and **99.9% pipeline reliability**.

TECHNICAL SKILLS

Languages & Databases: Python, SQL, Java, R, PostgreSQL (pgvector), SQLite, SQL Server, MongoDB, Snowflake

AI Applications & Agents: LLMs (GPT-4, Claude, Gemini, Llama 3), Agentic AI, RAG, Embeddings, Prompt Engineering, Structured Outputs, LangChain, LlamaIndex, Vector Databases (ChromaDB, Pinecone, FAISS), Azure OpenAI, Ollama

AI Testing & Evaluation: RAGAS, Recall@K, Hallucination Detection, Guardrails, Ablation Studies, A/B Testing

Back End & APIs: FastAPI, REST APIs, Streamlit, OAuth 2.0, API Versioning

DevOps & Cloud: Git/GitHub, GitHub Actions, CI/CD (Jenkins), Docker, Kubernetes, Infrastructure as Code (Terraform), AWS (S3, Lambda, SageMaker, SQS)

ML & Data: PyTorch, Scikit-learn, Hugging Face, BERT, XGBoost, MLflow, Apache Airflow, Spark/PySpark, Databricks

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, India

PROFESSIONAL EXPERIENCE

George Mason University | ***AI & ML Research Associate*** **Fairfax, VA | August 2024 - Present**

- To scale 9-1-1 call-taker training beyond instructor role-play, co-developed **SHIELD** with **Fairfax County DPSC**, where **LLM agents** simulate calls and coach trainees in real time; published at **ACM WSDM '26**.
- To take SHIELD from prototype to pilot, built its front end and back end in **Python/Streamlit** with **SQLite** accounts, course enrollment, and session tracking, iterating on instructor and agency feedback.
- To generate safe training calls without commercial APIs, co-developed **MATE-911**, a **RAG** application on **Llama 3.1 8B** reaching **0.81 protocol F1** and a **3.0% hallucination rate**; published at **IEEE PST 2026**.
- To give the agents trustworthy knowledge, led the **RAG data pipeline** turning unstructured call records into **160 gold-standard scenarios**, cutting hallucinations from **13.0% to 3.0%**.
- To ground agent responses, built **semantic search** with **embeddings**, **ChromaDB**, and metadata-filtered **top-K retrieval**, then used **ablation studies** to show it lifted the pass rate from **49.7% to 97.6%**.
- To keep live agent conversations accurate and fast, designed **prompts and AI workflows** for **Llama 3.1 8B**, delivering responses in under **1 second** for text and **1.3 seconds** for voice.
- To stop unsafe agent outputs before they reached trainees, led the **validation layer** with **protocol guardrails** and **hallucination checks**, lowering hallucinations from **9.5% to 3.0%**.
- To speed up faculty and student research, built a **RAG research assistant** with **LlamaIndex**, **pgvector**, and a **FastAPI** back end, raising **recall@10 from 61% to 88%** for **40+ researchers** at **190 ms p95**.
- To pick the right model for each task, integrated **GPT-4, Claude,** and **Gemini** through **LangChain** with **structured outputs**, raising answer accuracy by **16%** and cutting hallucinations by **19%**.
- To run LLMs on limited hardware, fine-tuned **Llama 3** with **LoRA/PEFT** and **quantization-aware training**, shrinking models by **40%** while staying within **3%** of full-precision accuracy.
- To ship AI services reliably, set up **MLflow**, **Docker**, **Git**, and **CI/CD** pipelines, cutting release cycles from **2 weeks to 5 days** with **A/B testing** of model and prompt versions.

Carelon Global Solutions (Elevance Health) | ***Machine Learning Engineer*** **Bengaluru, India | May 2022 - August 2023**

- To speed up member support, built a **Generative AI** assistant with **Azure OpenAI**, **LangChain**, and **Pinecone** RAG for **15K+ daily queries**, cutting resolution time by **38%** and raising ticket deflection by **25%**.
- To automate clinical text review, built **BERT/BioBERT** extraction models on unstructured medical notes, reaching **80% accuracy** and cutting processing time by **55%**.
- To integrate AI with clinical systems, containerized model services with **Docker, Kubernetes,** and **Jenkins CI/CD** behind **REST APIs**, improving uptime by **20%** and enabling real-time EHR integration.
- To keep production models accurate over time, set up **monitoring**, **drift detection**, and automated retraining, sustaining **over 95% accuracy** for **12 months** across **200+ end users**.
- To feed AI applications reliable data, designed **20+ pipelines** with **Airflow, PySpark,** and **AWS (S3, Lambda, Glue)**, raising throughput by **70%** and reaching **99.9% reliability** under **HIPAA**.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - April 2022**

- To secure data exchange between internal systems, developed **RESTful microservices** with **OAuth 2.0** authentication and **API versioning**, backed by **MongoDB** data models.
- To reduce manual claims reconciliation, redesigned **claims-balancing workflows** in **Python** and **SQL**, cutting operational overhead by **60%**.
- To eliminate costly invoice errors, automated **medical invoice processing** with rule-based classification, reducing manual errors by **70%**.
- To handle growing data volumes, migrated **Hive** and **Hadoop** workloads to **Apache Spark** and **Azure Data Services**, improving processing speed and scalability.

PROJECTS

Agentic AI Ticket Automation System | SecureGPT, LLaMA, AWS SQS, JIRA, ServiceNow

- To cut security-ticket triage time, built an **agentic AI** system with **SecureGPT** and **LLaMA** that monitors threats and auto-creates **JIRA** tickets under **NIST-CSF**, reaching **93% automation accuracy** and **70%** faster triage.
- To coordinate work across tools, simulated a **cross-agent** ticket lifecycle across **JIRA** and **ServiceNow** on **AWS SQS** event queues, using **prompt engineering** to rank events and improve scalability by **50%**.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, K. B. Laskey, L. Wells II, H. Purohit. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." *ACM WSDM '26*. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." *IEEE PST 2026*.